

Click-based Styling: Interactive Segmented Style Transfer

Johnny Qin, Yu Peng, Christopher Yeh, Yu-Ju Chiu
University of Michigan

{leqin, yupeng, chrisyeh, yujuchiu}@umich.edu

1. Introduction

Interactive segmentation has been researched for a while and various forms of interaction methods have been invented. For instance, users can provide bounding boxes [18], scribbles [12], or clicks [20] to extract style signals from images. These problems can be framed as a *mask extraction process*, where an algorithm receives input from users and outputs corresponding features for style transfer. Many systems such as GrabCut [18] and FocalClick [4] have tried to understand users' intention of scopes within clicks.

There has been also a rise in research towards neural style transfer that has explored a variety of directions, including optimization-based strategies, feed-forward models, and meta-learning techniques. A hybrid approach that combines per-pixel and perceptual loss functions is used to improve the performance toward real-time executions [11]. Besides that, more models use instance normalization to match the mean and variance of the style image to the content image. Quantitative methods have also been developed such as ArtFID [17] to support the automatic comparison of different style transfer approaches. However, the tools and software to connect click-based segmentation with style transfer is still lagging behind. To support content creators to express themselves freely, we propose a pipeline to complement this scheme in a timely manner. We provide extensive evaluations based on large datasets to examine the models we chose and their effects.

2. Related work

2.1. FID Fréchet Inception Distance

Heusel et al. [9] proposed the *Fréchet Inception Distance* (FID) to measure the *Fréchet Distance* [5] between the Gaussian distribution of Inception features of generated images and those of real images.

2.2. ArtFID [17]

Limited by the fact that style transfer lacks quantitative measurements for generated images and using human evaluation studies to assess stylization performance, Wright

and Ommer [17] propose a method to quantitatively evaluate style transfer models with respect to stylization performance that strongly coincides with human judgment. It combines the measure of content preservation and style matching. Style matching is measured by training a classifier on a large-scale artwork dataset and using the inception features from the last layer before the classifier heads as image representations. The distance between the feature distributions of the style images and the stylized images is then measured using FID [9]. In this case, only models that perform well on both content preservation and style matching achieve a low ArtFID score.

2.3. AdaIN [10]

Adaptive Instance Normalization is brought by Huang and Ulyanov [10, 22] to approximate the optimization problem formulated by Gatys et al. [6] for a specific style image. The basic idea behind AdaIN is to use instance normalization to match the mean and variance of the style image to the content image. The resulting normalized content image's feature maps are then scaled and shifted using the style image's mean and variance.

2.4. AdaAttN [15]

The Adaptive Attention Normalization model performs attentive normalization on a per-point basis and its weighed statistics are calculated by regarding a style feature point as a distribution of the attention-weighted output of all style feature points [15]. Additionally, the model is also enhanced by improving local visual quality through a novel local feature loss based on the derivation from its own model.

2.5. PAMA [16]

The Progressive Attentional Manifold Alignment algorithm introduces an enhancement toward arbitrary style transfer [16]. Specifically, the style transfer model is designed to align the content manifold to the style manifold, therefore alleviating the style degradation problem and producing state-of-the-art stylization results. Additionally, the model also consists of a channel alignment module to emphasize related content and style semantics, an attention

module to establish the correspondence between features, and a spatial interpolation module to adaptively align the manifolds [16].

2.6. Interactive segmentation

Interactive segmentation is taken as an optimization procedure before deep learning tools [7]. DIOS [23] first uses deep learning approaches to do interactive segmentation by adding positive and negative clicks into distance maps and concatenating them with the original image as input. After this, many other methods follow the same pipeline. [13, 14] Later, CDNet [3] introduces a self-attention mechanism into click-based interactive segmentation to generate more consistent results. RITM [21] applied the previous mask as network input to make the prediction more accurate.

2.7. FocalClick

Although excellent performance has been achieved in previous methods, their inference speed and preexisting masks still remain in the space of improvement. FocalClick is proposed by Chen et al. [4] to solve this problem. Local inference for interactive segmentation is brought, to leverage the Target Crop to the Focus Crop area, enabling resizing the Target Crop to small scales for higher speed.

3. Methods

In this model, the pipeline of networks consists of a Conditional Diffusion Network (CDNet) [3], an image transformation network [11], a loss network [11], and an AdaIN style transfer step, whereas each part focuses on object segmentation, image transformation, feature reconstruction loss for deep learning training purposes respectively.

3.1. Conditional Diffusion Network [3]

The pipeline of FocalClick is shown in Fig. 1 in which positive and negative clicks are embedded as two Gaussian maps. The information from the Gaussian maps and the image is concatenated to get a 5-channel input, whereas the input is then fed into a segmentation network to extract high-level features. In this network, we use DeeplabV3+ [2] with ResNet-50 [8] backbone with identical parameters similar to the original network from the reference. After the input information is been extracted through the segmentation model, the high-level features are fed into a Feature Diffusion Module (FDM) and a Pixel Diffusion Module (PDM) for predictions.

3.2. Image Transformation Network [11]

The image transformation network is a deep residual convolutions neural network parameterized by weights W . This is trained using Adam to minimize a weighted combi-

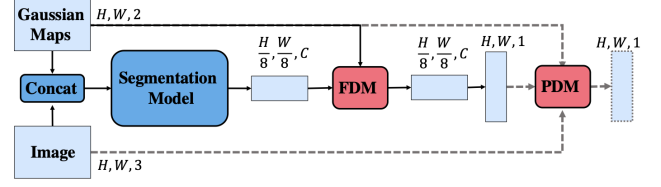


Figure 1. Conditional Diffusion Network overview [3].

nation of loss functions [11]:

$$W^* = \arg \min_W \mathbf{E}_{x, \{y_i\}} \left[\sum_{i=1} \lambda_i \ell_i (f_W(x), y_i) \right] \quad (1)$$

In the above expression, the target image is denoted as y_i , and several loss functions are denoted as $\ell_1, \dots, \ell_k$.

3.3. Loss Network [11]

The loss network is defined as a feature reconstruction loss ℓ_{feat}^ϕ and a style reconstruction loss ℓ_{style}^ϕ that measure differences in content and style between images [11]. For style transfer, we have a content target y_c as the input image x , and the output image $\hat{y}$ as the combined content of x with the style of y_s . For super-resolution, the input image is a low-resolution input, and content target y_c is the ground-truth high-resolution image.

4. Experiments

We selected 5 images of each category at random from the Wikiart dataset as our experiment subject. Then we used ClickSEG to obtain a mask of the main object in each image. To compare the relative performance between the selected models, we performed style transfer on the 585 objects from DAVIS585 with 135 style images selected previously to obtain a dataset of 78,975 images for each model. We selected a set of 1000 inputs (image, mask, style) at random and systematically retrieved corresponding stylized images, and evaluated these subsets using ArtFID.

To apply segmentation to our images, we adapted the selected model to combine stylized selected objects using masks created in the previous step and combine them with the background utilizing the content image and the inverse of the mask. We adapted from each model because of size deviations caused by stylization such as padding in convolution layers.

To evaluate the result of each model utilizing ArtFid, we resized the output from style transfer to 256x256.

Besides evaluating the stylization performance of these models, we also measured the time required to perform stylization by each model to compare the efficiencies. We computed the average time required to stylize one image by averaging 600 style transfers for each model. We only included the time required from style transfer to eliminate the

impact of subsequent steps such as applying masks to content and stylized images.

Style Transfer Algorithm	ArtFID $\uparrow$	runtime (ms)
AdaAttN	27.419	37.3597
PAMA	27.541	9.2811
IEContraAST	27.946	10.1938
AdaIN	27.955	4.9166

Table 1. Top 4 models with the smallest ArtFID [17] value. The lower ArtFID value indicates better performance. Experiments are conducted on 1000 images randomly selected from DAVIS585.

4.1. Use of Datasets

The above experiments including content images and style images are executed based on the following datasets: DAVIS585 [4] and WikiArt [19] that are originally introduced for interactive image segmentation and fine-art classification purposes respectively. DAVIS585 contains images and the corresponding segmentation masks from various categories, whereas the mask images identify the segmented objects from their original content images throughout all categories [4]. In this case, multiple objects may be detected or chosen by the user on one specific image; therefore, each content image may also contain several mask images which specify the different segmented objects. With this in mind, DAVIS585 not only allows us to get access through various types of content images but also provides mask information for segmented style transfer purposes. For experimental purposes, all images with their "first" corresponding mask images are utilized throughout the experiment, in which each pair of content and mask images are further used for segmented style transfer.

WikiART consists of artistic paintings that originated across 195 different artists, whereas the dataset is introduced for large-scale fine-art style classification [19]. For this purpose, 270 images are chosen of which 27 different styles are examined with 10 images each, in which all style images are performed toward segmented style transfer throughout the content images selected above. With the chosen content images and style images from DAVIS585 and WikiART, these images are stylized and generated through the listed style transfer algorithms including AdaIN [10], AdaAttN [15], PAMA [16], and IEContraAST [1], in which these constructed images are then evaluated according to their original content images and the provided style images through ArtFID, a quantitative evaluation method for style transferred images [17].

4.2. Evaluations

Our research is original research, and there are few other studies that combine segmentation and style transfer. To evaluate the results of style transfer for objects, we measure ArtFID scores from segmentation style transfer images. We use our dataset Davis585 and WikiArt to generate 1000 stylized images for each style transfer algorithm (Each image with at least one object) and use ArtFID to calculate scores from these 1000 images and get the average. The score is about 25, which is in line with the trend of ArtFID scores. The following are our ArtFID scores for four different Style Transfers Algorithms.

5. Conclusions

In this work, we propose a method for interactively style transfer by proposing a pipeline connecting interactive segmentation with a style transfer model. By referencing extensive literature reviews and experiments, we selected FocalClick [4] for its excellent performance within a timely manner and AdaIN for its lowest ArtFID value. The goal of this work is to provide content creators with a tool to experiment with different art styles on different parts of a content image. Taking advantage of the fast execution time of FocalClick [4] and AdaIN [10], users can achieve real-time responses from both clicks and generation, allowing them have more time to experiment with their works. We anticipate that stylizing part of the image to the user's preference can pave the way to more creative images and videos, appealing works of digital art, or advertisement with products stylized.

5.1. Novelty

The combination of image segmentation and style transfer has been made possible in the past with the sacrifice of time efficiency which makes it hard to be applied in real-time applications. With segmentation and style transfer models available recently, we created a model of pipeline allowing real-time style transfer from user-selected segmentation within clicks. Furthermore, our model has many potential applications in the real world such as streaming for artists and content creators with no need for post-processing. Additionally, it could also be applied to video conferencing and only apply style transfer to a particular section of the video. In brief, it generates countless possibilities for artistic tasks including content creation, art entertainment, virtual reality, and many more.

5.2. Limitation

Due to the hardness of the quantitative measure of the quality of generated images, ArtFID is the only measure we selected to pick the best model for our pipeline to use. Due to time restrictions, we didn't carry out volunteer-based

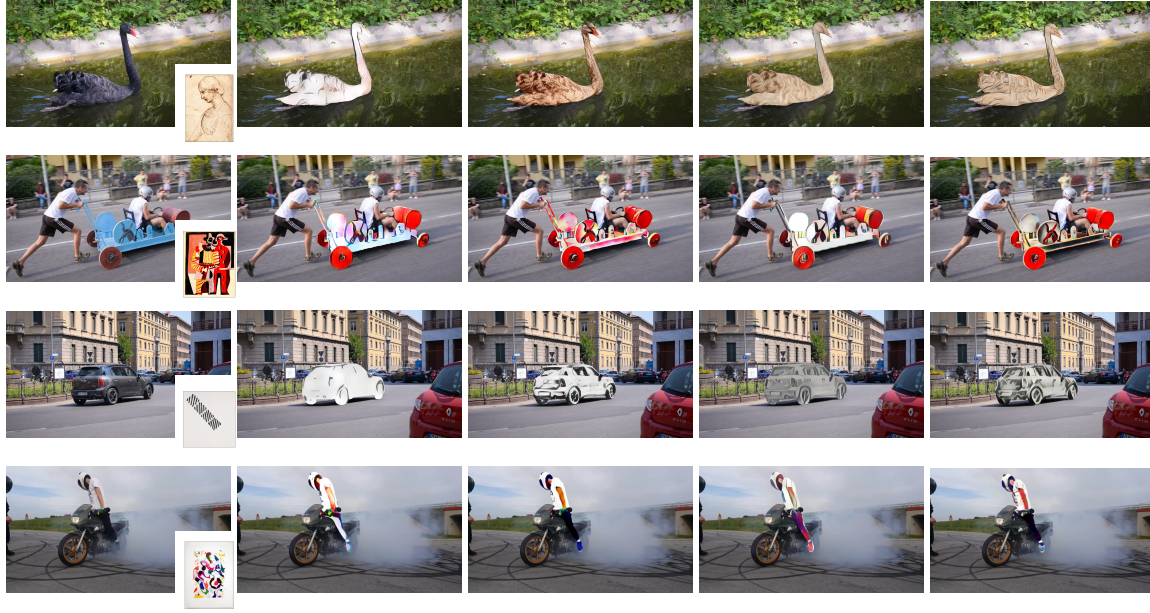


Figure 2. Result of performing segmented style transfer on DAVIS585 dataset. From the left to right we have the content image with its style image, segmented stylized images from AdaIn, IEContraAST, PAMA, and AdaAttN.

user study to see if qualitative measurements can align with ArtFID values.

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